

Notes for Section 2.3 of Lazarsfeld's Positivity in Algebraic Geometry I

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1 Zariski's Construction

$C_0 \subseteq \mathbb{P}^2$ is a smooth cubic curve, $l \in |\mathcal{O}_{\mathbb{P}^2}(1)|$. Notice that the genus g of C_0 is:

$$g = \frac{(\deg C_0 - 1)(\deg C_0 - 2)}{2} = 1.$$

So C_0 is an elliptic curve.

By Bezout's Lemma:

$$\deg \mathcal{O}_{C_0}(l) = (l \cdot C_0) = 3.$$

Choose arbitrary 12 points $p_1, \dots, p_{12} \in C_0$, then:

$$\deg \mathcal{O}_{C_0}(p_1 + \dots + p_{12} - 4l) = 0.$$

i.e. the equivalent class of $\mathcal{O}_{C_0}(p_1 + \dots + p_{12} - 4l)$ in $\text{Pic}(C_0)$ is in $\text{Pic}^0(C_0)$. We denote the line bundle by η .

We can further **require** that $[\eta]$ is not a torsion element in $\text{Pic}^0(C_0)$. There is a well-known property of elliptic curves that there is a bijection:

$$\text{Pic}^0(C_0) \leftrightarrow C_0(\mathbb{C}).$$

And by $C_0(\mathbb{C}) \simeq \mathbb{C}/\Lambda$ for some lattice Λ ,

$$\text{Tor}(\text{Pic}^0(C_0)) \text{ is countable.}$$

By the uncountability of $C_0(\mathbb{C})$, we can choose $[\eta]$ to be a non-torsion element in $\text{Pic}^0(C_0)$.

Now we blow up $\mathbb{P}^2$ at $p_1, \dots, p_{12}$:

$$\mu : X := \text{Bl}_{\{p_1, \dots, p_{12}\}} \rightarrow \mathbb{P}^2.$$

The exceptional divisor is denoted by:

$$E = \sum_{i=1}^{12} E_i \text{ with each } E_i \cong \mathbb{P}^1.$$

Let $H = \mu^*l$, $C \in |3H - E|$ be the strict transform of C_0 . Let $D = 4H - E$. Notice that $D \equiv_{\text{lin}} C + H$, where C is an effective divisor and H is a big divisor by [Lazarsfeld] Proposition 2.2.43 (Birational invariance of volume), so D is big.

We can prove D is nef. Notice that:

$$\begin{aligned}\mathcal{O}_C(D) &= \mathcal{O}_C(4H) \otimes \mathcal{O}_C(-E) \\ &\cong \mathcal{O}_{C_0}(4l) \otimes \mathcal{O}_{C_0}(-p_1 - \dots - p_{12}) \\ &\cong \eta^\vee,\end{aligned}$$

which means:

$$(D \cdot C) = \deg \mathcal{O}_C(D) = \deg \eta^\vee = 0.$$

For any irreducible curve $C' \subseteq \text{Bl}_{\{p_1, \dots, p_{12}\}}$ with $C' \neq C$, we have:

$$(D \cdot C') = (C \cdot C') + (H \cdot C') > 0.$$

So by definition of nefness, D is nef.

We can prove that C is contained in the base locus of $|mD|$ for any $m \in \mathbb{Z}_{>0}$. It suffices to show the following homomorphism is the zero map:

$$H^0(X, \mathcal{O}_X(mD)) \rightarrow H^0(C, \mathcal{O}_C(mD)).$$

In fact, $H^0(C, \mathcal{O}_C(mD)) = 0$, since otherwise with $\deg \mathcal{O}_C(mD) = -m \cdot \deg \eta = 0$, $\mathcal{O}_C(mD)$ can only be trivial, which means $[\eta]$ is a torsion element in $\text{Pic}^0(C_0)$, contradicting to our choice of $[\eta]$.

We can prove by induction that $|mD - C|$ is base point free for any $m \in \mathbb{Z}_{>0}$. When $m = 1$, $|D - C| = |H|$ is base point free because the pull back of a globally generated line bundle is globally generated. If we proved $|(m-1)D - C|$ is base point free, consider the following exact sequence:

$$0 \rightarrow \mathcal{O}_X((m-1)D + H - C) \rightarrow \mathcal{O}_X((m-1)D + H) \cong \mathcal{O}_X(mD - C) \rightarrow \mathcal{O}_C((m-1)D - H) \rightarrow 0.$$

We claim that:

$$H^1(X, \mathcal{O}_X((m-1)D + H - C)) = 0.$$

One quick but advancing way is using Kawamata-Viehweg Vanishing Theorem ([Lazarsfeld] Theorem 4.3.1.). By [Hartshorne] Exercise III.8.5:

$$K_X = \mu^* K_{\mathbb{P}^2} + E = -3H + E.$$

Notice that:

$$\begin{aligned} (m-1)D + H - C - K_X &\equiv_{\text{lin}} (m-1)D + H - (D - H) + 3H - (4H - D) \\ &= (m-1)D + H \end{aligned}$$

Both D and H are nef¹ and big, so $(m-1)D + H$ is nef and big. By Kawamata-Viehweg Vanishing Theorem, we have:

$$H^1(X, \mathcal{O}_X((m-1)D + H - C)) \cong H^1(X, \mathcal{O}_X(K_X + (m-1)D + H)) = 0.$$

Then we get the following commutative diagram for all $x \in X$:

$$\begin{array}{ccccccc} 0 & \longrightarrow & H^0(X, \mathcal{O}_X(M - C)) & \longrightarrow & H^0(X, \mathcal{O}_X(M)) & \longrightarrow & H^0(C, \mathcal{O}_C(M)) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathcal{O}_X(M - C)_x & \longrightarrow & \mathcal{O}_X(M)_x & \longrightarrow & \mathcal{O}_C(M)_x \longrightarrow 0 \end{array}$$

where $M := (m-1)D + H$. The first vertical map is surjective by $|(m-1)D - C|$ and $|H|$ being base point free. The third vertical map is surjective because:

$$\deg \mathcal{O}_C(M) = ((m-1)D \cdot C) + (H \cdot C) = 0 + 3 \geq 2g = 2,$$

and by [Hartshorne] Chapter IV Corollary 3.2., $\mathcal{O}_C(M)$ is globally generated. So by 5-Lemma, the second vertical map is surjective, which means $|mD - C|$ is base point free.

We conclude what we have proved so far:

- $C \subseteq \text{Bs}(|mD|)$ for any $m \in \mathbb{Z}_{>0}$.
- $|mD - C|$ is base point free for any $m \in \mathbb{Z}_{>0}$.

In geometric sense, this means that for all $m \in \mathbb{Z}_{>0}$, we can always find a divisor $D_m \in |mD|$ such that $D_m = C + E_m$ with E_m being an effective divisor and C not being any component of E_m .

Now we consider the following graded ring:

$$R(X, D) := \bigoplus_{m \geq 0} H^0(X, \mathcal{O}_X(mD)).$$

Using the mentioned geometric sense, we can prove that $R(X, D)$ is not finitely generated $\mathbb{C}$ -algebra. Suppose $R(X, D)$ is finitely generated by $r_1, \dots, r_k$ with $r_j \in H^0(X, \mathcal{O}_X(m_j D))$ for

¹Nefness of H comes from the fact that proper pull back of a nef divisor is still nef.

some $m_j \in \mathbb{Z}_{>0}$. For $m \gg 0$, any element in $H^0(X, \mathcal{O}_X(mD))$ can be written as:

$$\sum_{\sum_j \alpha_j m_j = m} c_j r_1^{\alpha_1} \cdots r_k^{\alpha_k}.$$

Consider the multiplicity of C in the divisor corresponding to this element. The multiplicity must be at least $\min_{\sum_j \alpha_j m_j = m} (\sum_j \alpha_j) \geq \frac{m}{\max_j (m_j)}$, which contradicts to the existence of D_m for $m \gg 0$.

What we have done is:

Theorem 1.1. There exists a smooth surface X and a nef and big divisor D on X such that the graded ring $R(X, D)$ is not finitely generated as a $\mathbb{C}$ -algebra.

By imitating the above construction, we now construct² another surface X with certain condition and a big divisor D on X such that the graded ring $R(X, D)$ is not finitely generated as a $\mathbb{C}$ -algebra.

Example 1.2. Let X be a projective surface, C be an irreducible curve on X such that

- (i) $(C^2) < 0$.
- (ii) $\text{Pic}(X) \rightarrow \text{Pic}(C)$ is injective.

Choose an ample divisor A on X . Let $D = (C \cdot A)C - (C \cdot C)A$. Then $R(X, D)$ is not finitely generated as a $\mathbb{C}$ -algebra.

Proof Sketch. Denote $a := (C \cdot A)$ and $b := -(C^2)$, then $a, b \in \mathbb{Z}_{>0}$. Notice that $(D \cdot C) = 0$, so D is nef. Bigness is also immediate. $C \subseteq \text{Bs}(|mD|)$ for any $m \in \mathbb{Z}_{>0}$ comes from $(D \cdot C) = 0$ and $\mathcal{O}_C(D)$ is not a torsion element in $\text{Pic}(C)$ ³.

A different point is that we now construct $p_0 \in \mathbb{Z}_{>0}$ such that $|mD - p_0 C|$ is base point free for any $m \gg 0$ instead of $|mD - C|$. By Castelnuovo-Mumford Regularity, it suffices to require that:

$$\begin{aligned} H^1(X, \mathcal{O}_X(mD - p_0 C - A)) &= 0, \forall m \in \mathbb{Z}_{>0}. \\ H^2(X, \mathcal{O}_X(mD - p_0 C - 2A)) &= 0, \forall m \in \mathbb{Z}_{>0}. \end{aligned}$$

²See [Laz] Example 2.3.1.

³Otherwise by the injectivity, there exists $m \in \mathbb{Z}_{>0}$ such that $mD \equiv_{\text{lin}} 0$, contradicting the bigness of D .

Notice that:

$$mD - p_0C - 2A = (bk - 2)A + (ak - p_0)C + (m - k)D, \forall m, k \in \mathbb{Z}_{>0}.$$

By Fujita Vanishing Theorem, there exists $m_0 \in \mathbb{Z}_{>0}$ such that:

$$H^i(X, \mathcal{O}_X(mA + N)) = 0, \forall m \geq m_0, \forall i > 0, \forall N \text{ nef.}$$

Choose $k, p_0 \in \mathbb{Z}_{>0}$ such that $ak - p_0 = 0$ and $bk - 2 \geq m_0$. By the nefness of D , the required vanishings hold, so $|mD - p_0C|$ is base point free for any $m \gg 0$.

Suppose $R(X, D)$ is finitely generated, analyzing the multiplicity of C in the divisor corresponding to an element in $H^0(X, \mathcal{O}_X(mD))$ for $m \gg 0$, we can get a contradiction. $\square$

2 Cutkosky's Construction

Let V be an irreducible projective variety with $\dim V = v$. Let $A_0, \dots, A_r$ be integral Cartier divisors on V . Put

$$\mathcal{E} := \bigoplus_{i=0}^r \mathcal{O}_V(A_i),$$

and consider the projective bundle $X := \mathbb{P}(\mathcal{E}) \rightarrow V$ with X being $n = v + r$ -dimensional irreducible projective variety.

Lemma 2.1. 1. $H^0(X, \mathcal{O}_X(k)) = \bigoplus_{\substack{\sum_i \alpha_i = k \\ \alpha_i \geq 0}} H^0(V, \mathcal{O}_V(\alpha_0 A_0 + \dots + \alpha_r A_r)).$

2. $\mathcal{O}_X(1)$ is ample if and only if A_i is ample for any i .

3. $\mathcal{O}_X(1)$ is nef if and only if A_i is nef for any i .

4. $\mathcal{O}_X(1)$ is big if and only if there exist $\alpha_i \geq 0$ such that $0 \neq \alpha_0 A_0 + \dots + \alpha_r A_r$ is big.

5. $\mathcal{O}_X(m)$ is globally generated for some $m \in \mathbb{Z}_{>0}$ if and only if $\mathcal{O}_V(m A_i)$ is globally generated for any i .

Proof of (4). See [Lazarsfeld] Lemma 2.3.2.

If there is some non-negative integers α_j such that $0 \neq \alpha_0 A_0 + \dots + \alpha_r A_r$ is big, consider the linear map:

$$\begin{aligned} \varphi : \mathbb{R}^{r+1} &\longrightarrow N^1(V)_{\mathbb{R}} \\ (x_0, \dots, x_r) &\longmapsto x_0[A_0] + \dots + x_r[A_r]. \end{aligned}$$

The condition means:

$$(\mathbb{Z}_{\geq 0}^{r+1} \cap \varphi^{-1}(\text{Big}(V))) \setminus \{0\} \neq \emptyset.$$

Because $\text{Big}(X)$ is an open cone, we can choose a basis $\{B_j\}$ of $\mathbb{R}^{r+1}$ in $\mathbb{Z}_{\geq 0}^{r+1} \cap \varphi^{-1}(\text{Big}(V))$.

Fix an ample divisor H on V . Then there exists $m_j \in \mathbb{Z}_{>0}$, E_j being effective divisors such

that $m_j B_j \equiv_{\text{lin}} E_j + H$ by the bigness of B_j . Write $B_j = \sum_{i=0}^r b_{ij} A_i$ with $b_{ij} \in \mathbb{Z}_{\geq 0}$. Then:

$$\begin{aligned}
H^0(X, \mathcal{O}_X(m)) &= \bigoplus_{\substack{\sum_i \alpha_i = m \\ \alpha_i \geq 0}} H^0(V, \mathcal{O}_V(\sum_{i=0}^r \alpha_i A_i)) \\
&\geq \bigoplus_{\substack{\sum_j m_j b_{ij} \cdot \beta_j = m \\ \beta_j \geq 0}} H^0(V, \mathcal{O}_V(\sum_{i=0}^r \beta_j \cdot m_j B_j)) \\
&\geq \bigoplus_{\substack{\sum_j m_j b_{ij} \cdot \beta_j = m \\ \beta_j \geq 0}} H^0(V, \mathcal{O}_V(\sum_{i=0}^r \beta_j H)) \\
&\geq \# \left\{ (\beta_0, \dots, \beta_r) \mid \beta_j \geq 0, \sum_j m_j b_{ij} \cdot \beta_j = m \right\} \cdot C \cdot \left(\frac{r \cdot m}{\max_j (m_j b_{ij})} \right)^v
\end{aligned}$$

There is a fact that the number of non-negative integer solutions of $\sum_j m_j b_{ij} \cdot \beta_j = m$ is asymptotically $C' \cdot m^r$ for some constant $C' > 0$. Hence $\dim H^0(X, \mathcal{O}_X(m)) \geq C'' \cdot m^n$ for some constant $C'' > 0$, which means $\mathcal{O}_X(1)$ is big.

Conversely, if $\mathcal{O}_X(1)$ is big, we prove by contradiction. Suppose for all $a_0, \dots, a_r \in \mathbb{N}$, $a_0 A_0 + \dots + a_r A_r$ are not big. We first claim that for all ample divisor H , $h^0(V, \mathcal{O}_V(a_0 A_0 + \dots + a_r A_r - H)) = 0$. It suffices⁴ to show that

$$a_0 A_0 + \dots + a_r A_r - H \notin \overline{\text{Eff}(V)}$$

Suppose $a_0 A_0 + \dots + a_r A_r - H \in \overline{\text{Eff}(V)}$. By openness of ampleness, there exists an open neighborhood U of H in $N^1(X)_{\mathbb{R}}$ such that $U \subseteq \text{Amp}(V) \subseteq \text{Big}(V) \subseteq \overline{\text{Eff}(V)}$. So $(a_0 A_0 + \dots + a_r A_r - H) + U$ is an open neighborhood of $a_0 A_0 + \dots + a_r A_r$ in $N^1(X)_{\mathbb{R}}$ which is contained in $\overline{\text{Eff}(V)}$. By $\text{Big}(V) = \text{Int}(\overline{\text{Eff}(V)})$, $a_0 A_0 + \dots + a_r A_r$ is big, which contradicts to our assumption.

Now choose a very ample divisor H and $m \in \mathbb{Z}_{>0}$ such that for all $i \in [0, r]$, $\mathcal{O}_H(mH - A_i)$ is very ample on H . $mH|_H - A_i|_H \sim N_i$ with N_i effective⁵. Consider

$$0 \rightarrow \mathcal{O}_V(a_0 A_0 + \dots + a_r A_r - H) \rightarrow \mathcal{O}_V(a_0 A_0 + \dots + a_r A_r) \rightarrow \mathcal{O}_H(a_0 A_0 + \dots + a_r A_r) \rightarrow 0$$

⁴If $a_0 A_0 + \dots + a_r A_r - H \notin \overline{\text{Eff}(V)}$ and simultaneously $h^0(V, \mathcal{O}_V(a_0 A_0 + \dots + a_r A_r - H)) \neq 0$, $\mathcal{O}_V(a_0 A_0 + \dots + a_r A_r - H)$ has at least one global section, which means $a_0 A_0 + \dots + a_r A_r - H \in \text{Eff}(V)$, a contradiction.

⁵**Notice:** When $\dim V \geq 2$, we can choose H to be integral, then the existence of N_i holds. If $\dim X = 1$, then $\dim H = 0$ and in the upcoming step, $h^0(V, \mathcal{O}_V(a_0 A_0 + \dots + a_r A_r)) \leq h^0(H, \mathcal{O}_H) = O(1) = O(k^0)$, which is still correct.

and we will get:

$$\begin{aligned}
h^0(V, \mathcal{O}_V(a_0A_0 + \dots + a_rA_r)) &\leq h^0(H, \mathcal{O}_H(a_0A_0 + \dots + a_rA_r)) + 0 \\
&\leq h^0(H, \mathcal{O}_H(-a_0N_0 - \dots - a_rN_r + kmH)) \\
&\leq h^0(H, \mathcal{O}_H(kmH)) \\
&\leq C_H m^{v-1} \cdot k^{v-1} \quad k \gg 0, \sum_i a_i = k.
\end{aligned}$$

So

$$\begin{aligned}
h^0(X, \mathcal{O}_X(k)) &= \sum_{a_0 + \dots + a_r = k} h^0(V, \mathcal{O}_V(a_0A_0 + \dots + a_rA_r)) \\
&\leq \binom{k+r}{k} C_H m^{v-1} \cdot k^{v-1} \leq C k^{n-1} \quad k \gg 0,
\end{aligned}$$

which contradicts the bigness of $\mathcal{O}_X(1)$. □

Theorem 2.2. Let V be an irreducible smooth projective curve with $g \geq 1$. Let $P \in \text{Pic}^0(V)$ and not be torsion. Let A be an ample divisor on V with a sufficiently large degree. Let $X := \mathbb{P}(\mathcal{O}_V(P) \oplus \mathcal{O}_V(A))$. Then $\mathcal{O}_X(1)$ is nef and big by 2.1, and

$$R := \bigoplus_{k \geq 0} H^0(X, \mathcal{O}_X(k))$$

is not finitely generated as a $\mathbb{C}$ -algebra.

Proof. See [Lazarsfeld] Example 2.3.3.

We claim that for $R_{i,j} := H^0(V, \mathcal{O}_V(iP + jA)) \subseteq H^0(X, \mathcal{O}_X(i+j)) = R_{i+j}$, $R_{k-1,1} \neq 0$ and it can not be generated by elements in lower degree. By $\deg((k-1)P + A) = \deg A$ is sufficiently large, $R_{k-1,1} \neq 0$. On the other hand,

$$\sum_{p=1}^{k-1} (R_{k-p-1,1} \otimes R_{p,0}) = 0$$

This is because by $\deg P = 0$ and P not being torsion, $R_{p,0} = 0$ for all $p \in \mathbb{Z}_{>0}$. □

See [Lazarsfeld] Example 2.3.5. Now let $g = 1$ and $A = x$, $\deg A = 1$.

Consider the section $\sigma : V \rightarrow X$ such that $\sigma^* \mathcal{O}_X(1) = \mathcal{O}_V(P)$.⁶ By the footnote, $\sigma(V)$

⁶You can understand it affine locally. For any affine open subscheme $\text{Spec } B$ of V , let $\mathcal{O}_V(P)|_{\text{Spec } B} = \widetilde{B \cdot s_0}$ and $\mathcal{O}_V(A)|_{\text{Spec } B} = \widetilde{B \cdot s_1}$. Then the affine local information of a section σ is $\text{Spec } B \rightarrow \text{Proj } B[s_0, s_1]$.

must be contained in the base locus the sublinear system $H^0(V, \mathcal{O}_V(A))$ of $H^0(X, \mathcal{O}_X(1))$. Then for all $k \in \mathbb{Z}_{>0}$ and $0 < i \leq k$, $\sigma(V)$ is contained in the base locus of the sublinear system $H^0(V, \mathcal{O}_V((k-i)P + iA))$ of $H^0(X, \mathcal{O}_X(k))$. Finally, since $\deg P = 0$ and P is not torsion, $H^0(V, \mathcal{O}_V(kP)) = 0$ for all $k \in \mathbb{Z}_{>0}$. So:

$$\sigma(V) \subseteq \text{Bs}(|\mathcal{O}_X(k)|), \quad \forall k \in \mathbb{Z}_{>0}.$$

We denote $\sigma(V)$ by F .

Notice that $\deg((k-1)P + A) = 1 > 2g - 2 = 0$. So $h^1(V, \mathcal{O}_V((k-1)P + A)) = h^0(V, \mathcal{O}_V(K_V - (k-1)P - A)) = 0$ for all $k \in \mathbb{Z}_{>0}$ and by Riemann-Roch formula,

$$h^0(V, \mathcal{O}_V((k-1)P + A)) = 1.$$

This means there exists a unique effective divisor y_k such that $y_k \equiv_{\text{lin}} (k-1)P + A$.

One question is what can be left after moving out F from every element in $|\mathcal{O}_X(k)|$. It suffices to analyze $|\mathcal{O}_X(k) \otimes \mathcal{O}_X(-F)|$. We restrict it on F :

$$\begin{aligned} \mathcal{O}_F(k) \otimes \mathcal{O}_F(-F) &= \sigma^* \mathcal{O}_X(k) \otimes (\sigma^* \mathcal{O}_X(1)^{\otimes 2} \otimes \det(\mathcal{O}_V(P) \oplus \mathcal{O}_V(A))^{\vee})^{\vee} \\ &= \mathcal{O}_V(kP - 2P + P + A) = \mathcal{O}_V((k-1)P + A). \end{aligned}$$

The first equation comes from [Vakil] 20.2.I. So after moving out F from every element in $|\mathcal{O}_X(k)|$ they must contain $z_k = \sigma(y_k)$.

Theorem 2.3. We can construct big line bundles on different varieties having arbitrarily small volume.

Proof. See [Lazarsfeld] Example 2.3.6.

V is an irreducible smooth projective curve with $g = 1$. $x \in V$ is a fixed closed point. For

Easy to see that the section σ_1 such that $\sigma_1^* \mathcal{O}_X(1) = \mathcal{O}_V(A)$ is just $V(s_0)$ (The global section $s_0 \in H^0(\text{Proj } B[s_0, s_1], \mathcal{O}(1))$ restricts to 0 on $V(s_0)$, while s_1 restricting to “itself” on $V(s_0)$.) And the section σ_2 such that $\sigma_2^* \mathcal{O}_X(1) = \mathcal{O}_V(P)$ is just $V(s_1)$.

any $a \in \mathbb{Z}_{>0}$, let $A_0 := (1 - a) \cdot x$, $A_1 := x$ and consider $X = \mathbb{P}(\mathcal{O}_V(A_0) \oplus \mathcal{O}_V(A_1))$. Then:

$$\begin{aligned} h^0(X, \mathcal{O}_X(k)) &= \sum_{i=0}^k h^0(V, \mathcal{O}_V(i \cdot x + (k - i)(1 - a) \cdot x)) \\ &= \sum_{i=0}^{\lfloor \frac{k}{a} \rfloor} (k - ia) + O(1) \\ &= \frac{1}{2a} k^2 + O(k). \end{aligned}$$

The second equation comes from $h^1(\deg > 0 = 2g - 2) = 0$ and Riemann-Roch formula. So $\text{vol}(\mathcal{O}_X(1)) = \frac{1}{a}$, which can be arbitrarily small. $\square$

Theorem 2.4. We can construct a big line bundle on a variety having irrational volume.

Proof. Consider $V := E \times E$ with E being an elliptic curve. For any two ample divisors D, H on V , consider

$$\sigma(D, H) := \max\{t \mid D - tH \text{ is nef}\}.$$

In fact, we can choose D and H such that $\sigma(D, H)$ is irrational. This is because by [Lazarsfeld] Lemma 1.5.4⁷, the boundary of $\text{Nef}(V)$ is defined by $(\delta^2) = 0$. So:

$$\sigma(D, H) = \text{smaller root of } ((D - tH)^2) = (D^2) - 2(D \cdot H) \cdot t + (H^2) \cdot t^2 = 0$$

For most choices of D and H , $\sigma = \sigma(D, H)$ is irrational.⁸

We claim that when $\frac{j}{i} < \sigma$:

$$h^0(V, \mathcal{O}_V(iD - jH)) = \frac{1}{2} ((iD - jH)^2).$$

By Riemann-Roch formula on surface:

$$\chi(V, \mathcal{O}_V(iD - jH)) = \frac{1}{2} ((iD - jH) \cdot (iD - jH - K_V)) + \chi(V, \mathcal{O}_V)$$

So it suffices to show:

⁷If you check the statement of Lemma 1.5.4, you can find there is another condition “ $(\delta \cdot h) \geq 0$ for some class h ”. We can ignore it because the nonnegative self-intersection condition gives us two components, “ample component” and “anti-ample component”. The latter condition is just used to draw out the correct part.

⁸We let σ be irrational because we want to avoid the case $\sigma = \frac{j}{i}$ for some $i, j \in \mathbb{Z}_{>0}$ in the following paragraph.

- $K_V = 0$.
- $\chi(V, \mathcal{O}_V) = 0$.
- $h^1(V, \mathcal{O}_V(iD - jH)) = h^2(V, \mathcal{O}_V(iD - jH)) = 0$.

Notice that:

$$\Omega_V^2 = \text{pr}_1^* \omega_E \oplus \text{pr}_2^* \omega_E = \mathcal{O}_E \oplus \mathcal{O}_E$$

by $\omega_E = \mathcal{O}_E$. So $\omega_V = \mathcal{O}_V$, i.e. $K_V = 0$. By Künneth formula:

$$H^k(V, \mathcal{O}_V) = \bigoplus_{i+j=k} H^i(E, \mathcal{O}_E) \otimes H^j(E, \mathcal{O}_E).$$

So by $h^0(E, \mathcal{O}_E) = 1$ and $h^1(E, \mathcal{O}_E) = 1$, $h^0(V, \mathcal{O}_V) = 1$, $h^1(V, \mathcal{O}_V) = 2$, $h^2(V, \mathcal{O}_V) = 1$ and $\chi(V, \mathcal{O}_V) = 0$. By the definition of σ , $iD - jH$ is an inner point of the nef cone when $\frac{j}{i} < \sigma$, hence ample. By Kodaira Vanishing Theorem, $h^i(V, \mathcal{O}_V(iD - jH)) = h^i(V, \mathcal{O}_V(K_V + iD - jH)) = 0$ for $i > 0$. So the claim is proved.

We can also find that when $\frac{j}{i} > \sigma$,

$$h^0(V, \mathcal{O}_V(iD - jH)) = 0.$$

This is because: suppose $h^0(V, \mathcal{O}_V(iD - jH)) \neq 0$, then $iD - jH$ is effective, hence $iD - jH \in \overline{\text{NE}}(V)$. But by [Lazarsfeld] Lemma 1.5.4, $\overline{\text{NE}}(V) = \text{Nef}(V)$, which contradicts to $\frac{j}{i} > \sigma$.

Consider $X = \mathbb{P}(\mathcal{O}_V(D) \oplus \mathcal{O}_V(-H))$. Then:

$$\begin{aligned} h^0(X, \mathcal{O}_X(k)) &= \sum_{\substack{i+j=k \\ \frac{j}{i} < \sigma}} h^0(V, \mathcal{O}_V(iD - jH)) \\ &= \sum_{\frac{k}{1+\sigma} < i \leq k} \frac{k^2}{2} \left(\left(\frac{i}{k} D - \left(1 - \frac{i}{k}\right) H \right)^2 \right) \end{aligned}$$

Hence:

$$\text{vol}(\mathcal{O}_X(1)) = \lim_{k \rightarrow \infty} \frac{h^0(X, \mathcal{O}_X(k))}{k^3/6} = 3 \cdot \int_{\frac{1}{1+\sigma}}^1 ((tD - (1-t)H)^2) dt,$$

which is irrational for most choices of D and H .

□

See [Lazarsfeld] Example 2.3.7. and again Example 2.3.5. The section $\sigma : V \rightarrow X$ such that $\sigma^* \mathcal{O}_X(1) = \mathcal{O}_V(-H)$ is just $E := \mathbb{P}(\mathcal{O}_V(-H)) \subseteq X$. We have discussed that E is contained in the base locus of the sublinear system $H^0(V, \mathcal{O}_V(D))$ of $H^0(X, \mathcal{O}_X(1))$. $-H$ is anti-ample, by Kodaira Vanishing Theorem, $H^0(V, \mathcal{O}_V(-kH)) = 0$ for all $k \in \mathbb{Z}_{>0}$. So E is contained in the base locus of $|\mathcal{O}_X(k)|$ for all $k \in \mathbb{Z}_{>0}$.

Now we want to analyze how big is the multiplicity of E in the base locus of $|\mathcal{O}_X(k)|$. By the previous discussion,

$$H^0(X, \mathcal{O}_X(k)) = \bigoplus_{\frac{k}{1+\sigma} < i \leq k} H^0(V, \mathcal{O}_V(iD - (k-i)H)).$$

The divisor D contributes 1 to the multiplicity of E in the base locus of $|\mathcal{O}_X(k)|$ and $-H$ contributes 0. So the multiplicity of E in the base locus of $|\mathcal{O}_X(k)|$ is $\left\lfloor \frac{k}{1+\sigma} \right\rfloor + 1$.

In fact we can show:

$$\mathfrak{b}(|\mathcal{O}_X(k)|) = \mathcal{O}_X \left(- \left(\left\lfloor \frac{k}{1+\sigma} \right\rfloor + 1 \right) E \right), \quad \forall k \gg 0.$$

It suffices to show set-theoretically $\text{Bs}(|\mathcal{O}_X(k)|) = E$. For any $x \in X \setminus E$, the image of x under the projection $\pi : X \rightarrow V$ is denoted by y . Because D is ample, for all $k \gg 0$, kD is very ample, then base point free. So there exists $s \in H^0(V, \mathcal{O}_V(kD))$ such that $s(y) \neq 0$. Let $\tilde{s} \in H^0(X, \mathcal{O}_X(k))$ which corresponds to s under the isomorphism $H^0(X, \mathcal{O}_X(k)) \cong \bigoplus_{\frac{k}{1+\sigma} < i \leq k} H^0(V, \mathcal{O}_V(iD - (k-i)H))$. x is in the fibre of y , i.e. $\mathbb{P}_{\kappa(y)}^1 = \text{Proj } \kappa(y)[T_0, T_1]$ with T_i corresponding to the basis of $\mathcal{O}_V(D)|_{\text{Spec } \kappa(y)}$ and $\mathcal{O}_V(-H)|_{\text{Spec } \kappa(y)}$ respectively. By $x \in X \setminus E$, see locally and we will find $x \in D_+(T_0)$. Then also see locally we will find:

$$\tilde{s}(x) = s(y) \cdot T_0^k(x) \neq 0.$$

So $x \notin \text{Bs}(|\mathcal{O}_X(k)|)$.

3 Base Loci of Nef and Big Linear Series

Theorem 3.1 (Wilson). Let X be irreducible projective variety with $\dim X = n$. Let D be a nef and big divisor on X . Then there exists $m_0 \in \mathbb{Z}_{>0}$ and an effective divisor N such that: for all $m \geq m_0$, we have

$$|mD - N| \text{ base point free.}$$

Proof. See [Lazarsfeld] Theorem 2.3.9.

By Fujita Vanishing Theorem, there exists a (very) ample B such that for all $i > 0$ and nef divisor P :

$$H^i(X, \mathcal{O}_X(B + P)) = 0.$$

By D being big, there exists $m_0 \in \mathbb{Z}_{>0}$ and an effective divisor N such that $m_0 D \equiv_{\text{lin}} (n + 1)B + N$. Then for all $m \geq m_0$:

$$\begin{aligned} mD - N &\equiv_{\text{lin}} (m - m_0)D + m_0 D - N \\ &\equiv_{\text{lin}} (n + 1)B + (m - m_0)D. \end{aligned}$$

Then by nefness of D and ampleness of B ,

$$\begin{aligned} H^i(X, \mathcal{O}_X(mD - N - iB)) &\cong H^i(X, \mathcal{O}_X((n + 1 - i)B + (\text{nef}))) \\ &= 0. \quad \forall i > 0, \forall m \geq m_0. \end{aligned}$$

By Castelnuovo-Mumford Regularity, $mD - N$ is base point free for all $m \geq m_0$. □

Remark. In Zariski's construction, N is taken to be C . While in fact, N not always of the form of $1 \cdot (\text{primeWeildivisor})$. For example, in [Lazarsfeld] Example 2.3.5, we have discussed that $|\mathcal{O}_X(k) \otimes \mathcal{F}|$ has a base point z_k , which means that taking $\mathcal{O}_X(D) = \mathcal{O}_X(1)$ and $N = F$ is wrong. We must at least take $N = 2F$. In fact, this is correct.

Definition 3.2 (Multiplicity of a linear series at a point). See [Lazarsfeld] Definition 2.3.11.

Let X be a projective variety and $x \in X$ a fixed point. D is a divisor on X and $|V| \subseteq |D|$ is linear series. We define:

$$\text{mult}_x |V| := \min_{D' \in |V|} \{\text{mult}_x D'\}$$

Corollary 3.3. See [Lazarsfeld] Corollary 2.3.12.

Let X be an irreducible projective variety and D a nef and big divisor on X . Then there exists $C > 0$ independent of m and x such that:

$$\text{mult}_x |mD| \leq C, \quad \forall x \in X, \forall m \in \mathbb{Z}_{>0}.$$

Proof. By the previous theorem, there exists $m_0 \in \mathbb{Z}_{>0}$ and an effective divisor N such that for all $m \geq m_0$, $|mD - N|$ is base point free. So

$$\text{mult}_x |mD - N| = 0, \quad \forall x \in X, \forall m \geq m_0.$$

Then for all $x \in X$ and $m \geq m_0$:

$$\text{mult}_x |mD| \leq \text{mult}_x |mD - N| + \text{mult}_x N = \text{mult}_x N.$$

We can take $C = \max_{x \in X} \{\text{mult}_x N\} + \max_{\substack{m \leq m_0 \\ x \in X}} \{\text{mult}_x |mD|\}$. □

Remark. In addition, Goodman tells us that if D satisfies that there exists $C > 0$ such that for all $x \in X$ and $m \in \mathbb{Z}_{>0}$, $\text{mult}_x |mD| \leq C$, then D is nef.

Remark. In our previous understanding, even if $D_1 \equiv_{\text{num}} D_2$, $\text{Bs}|D_1|$ and $\text{Bs}|D_2|$ can be different. However, Nakamaye tells us that if D is nef and big, then as $m \rightarrow \infty$, $\text{Bs}|mD|$ stabilizes to a fixed subset of X and it is independent of the choice of D in its numerical class. It is:

$$\bigcup_{(D^{\dim Y} \cdot Y)=0} Y,$$

and you can see it immediately when D is ample.

Remark. We can not remove bigness in the above theorem. The counterexample is the following: consider irreducible projective curve V with $g \geq 1$, and $X = \mathbb{P}(\mathcal{O}_V \oplus \mathcal{O}_V(P))$ where $P \in \text{Pic}^0(V)$ is a non-torsion. Then the section $\sigma : V \rightarrow X$ such that $\sigma^* \mathcal{O}_X(1) = \mathcal{O}_V(P)$ satisfies:

$$F := \sigma(V) \subseteq \text{Bs}|H^0(V, \mathcal{O}_V)|.$$

Notice that:

$$H^0(X, \mathcal{O}_X(m)) = \bigoplus_{k=0}^m H^0(V, \mathcal{O}_V(kP)) = H^0(V, \mathcal{O}_V) \quad \forall m \in \mathbb{Z}_{>0}$$

by the choice of P . So the multiplicity of F in $|\mathcal{O}_X(m)|$ is m , which means for all $x \in F$, $\text{mult}_x|\mathcal{O}_X(m)| = m$, being not bounded by a constant independent of m .

Theorem 3.4. Let X be a normal projective variety and D be nef and big divisor on X . Then $R(X, D)$ is finitely generated if and only if D is semi-ample.

Proof of ‘only if’. See [Lazarsfeld] Theorem 2.3.15.

If $R(X, D)$ is finitely generated by homogeneous elements $f_1, \dots, f_r$ with $\deg f_i = d_i$, Let $l = \text{lcm}(d_1, \dots, d_r)$. Then $H^0(X, \mathcal{O}_X(lD))$ generates $R(X, D)^{(l)} := \bigoplus_{k \in \mathbb{Z}_{\geq 0}} H^0(X, \mathcal{O}_X(klD))$. This means that for all $k \in \mathbb{Z}_{>0}$ and every section in $H^0(X, \mathcal{O}_X(klD))$, it can always be written as a ‘polynomial’ of sections in $H^0(X, \mathcal{O}_X(lD))$. Then we have:

$$\text{mult}_x|klD| \geq \text{mult}_x|lD| \cdot k, \quad \forall x \in X, \forall k \in \mathbb{Z}_{>0}.$$

On the other hand, because D is nef and big, there exists $C > 0$ such that $\text{mult}_x|mD| \leq C$ for all $x \in X$ and $m \in \mathbb{Z}_{>0}$. So $\text{mult}_x|lD| = 0$ for all $x \in X$. Then $|lD|$ is base point free and D is semi-ample. $\square$

4 Nakai criterion for $\mathbb{R}$ -divisors

Remark. *This important result can be read when learning volume of divisors. There is nothing else to add to the content in the book.*

Theorem 4.1. Let X be a projective scheme, and let $\delta \in N_1(X)_{\mathbb{R}}$ be a class such that for every irreducible subvariety $V \subset X$ we have

$$\delta^{\dim V} \cdot V > 0.$$

Then δ is ample.

Proof. There is no loss in generality in taking X reduced and irreducible, and the assertion is clear if $\dim X = 1$. So we proceed by induction on $n = \dim X$, and assume that for every proper subvariety $Y \subset X$ the restriction

$$\delta|_Y \in N^1(Y)_{\mathbb{R}}$$

is an ample class on Y . Note that in any event, δ is nef.

Figure 1: page 166.

Choose any norm on $N^1(X)_{\mathbf{R}}$ inducing the standard topology on that finite-dimensional real vector space. Arguing as in the proofs of 1.4.10 or 1.4.9, one sees that there exist arbitrarily small ample classes $\alpha, \alpha' \in N^1(X)_{\mathbf{R}}$ such that

$$\delta + \alpha' \quad , \quad \alpha + \alpha' \quad , \quad \text{and} \quad \delta - \alpha$$

are rational. Moreover since $(\delta^n) > 0$, we can suppose by taking α and α' sufficiently small that

$$\left((\delta + \alpha')^n \right) > n \cdot \left((\delta + \alpha')^{n-1} \cdot (\alpha + \alpha') \right).$$

It then follows from the numerical criterion for bigness (Theorem 2.2.15) that

$$\delta - \alpha = (\delta + \alpha') - (\alpha + \alpha')$$

is represented by an *effective* $\mathbf{Q}$ -divisor E . Denote by $Y_1, \dots, Y_t \subset X$ the irreducible components of a support of E .

We now claim that if $0 < \varepsilon \ll 1$ is sufficiently small, then $(\delta - \varepsilon \cdot \alpha)$ is a nef class on X . In fact $\delta|_{Y_i}$ is ample by the induction hypothesis. So by taking $0 < \varepsilon \ll 1$ we can arrange that each of the restrictions $(\delta - \varepsilon \cdot \alpha)|_{Y_i}$ are nef (Example 1.3.14). Now let $C \subset X$ be any curve. If $C \subset Y_i$ for some i , then $((\delta - \varepsilon \cdot \alpha) \cdot C) \geq 0$ by what we have just said. On the other hand, if $C \not\subset \text{Supp}(E)$, then evidently

$$(E \cdot C) = ((\delta - \alpha) \cdot C) \geq 0,$$

and a fortiori $((\delta - \varepsilon \cdot \alpha) \cdot C) \geq 0$. Thus $(\delta - \varepsilon \cdot \alpha)$ is indeed nef. But then δ — being the sum of a nef and an ample class — is ample. $\square$

5 Zariski Decomposition

Theorem 5.1 (Zariski Decomposition). Let X be a smooth projective surface, and let D be a pseudo-effective integral divisor on X . Then D can be written uniquely as a sum:

$$D = P + N$$

of $\mathbb{Q}$ -divisors with:

1. P is nef;
2. $N = \sum_{i=1}^q a_i C_i$ is effective, and if $N \neq 0$ then $\|(C_i \cdot C_j)\|_{i,j=1,\dots,q}$ is negative definite;
3. $(P \cdot C_i) = 0$ for all $i = 1, \dots, q$.

Remark. The divisor P (the positive part) is the “good” component. Geometrically, P captures the “mobile” portion of D . It is the well-behaved part of the divisor that carries the asymptotic global sections and dictates the growth of the linear system $|mD|$. The divisor N (the negative part) is the “bad” or rigid component. Geometrically, N is completely fixed on the surface. It behaves as a rigid base locus that cannot move, forcing negative intersections with the divisor.

We can see this from the following proposition.

Proposition 5.2. In the situation of Theorem 5.1, the natural map

$$H^0(X, \mathcal{O}_X(\lfloor mP \rfloor)) = H^0(\mathcal{O}_X(mD - \lceil mN \rceil)) \rightarrow H^0(X, \mathcal{O}_X(mD))$$

is an isomorphism for all $m \geq 0$.

Proof. Step 1: First prove the case when P, N are both integral divisors and $m = 1$. It suffices to show the following map is surjective (then automatically isomorphic):

$$H^0(X, \mathcal{O}_X(P)) \rightarrow H^0(X, \mathcal{O}_X(D)),$$

which is equivalent to showing the following map is surjective:

$$|P| \rightarrow |D| : E \mapsto E + N.$$

For any $D' \in |D|$, C_i as in Theorem 5.1, write:

$$D' = N_1 + M_1, \quad \text{where } N_1 \text{ is effective with components of } \{C_i\}_{i=1,\dots,q}, \\ \text{and } M_1 \text{ is effective with no components of } \{C_i\}_{i=1,\dots,q}.$$

Now we claim that $N_1 - N$ is effective. Write $N_1 - N = A - B$ where A and B are effective divisors with no common components. On one hand:

$$((N_1 - N) \cdot B) = (A \cdot B) - (B^2) \geq 0$$

where $(B^2) \leq 0$ by the negative definiteness of $\|(C_i \cdot C_j)\|_{i,j=1,\dots,q}$. On the other hand, for all $i = 1, \dots, q$:

$$((N_1 - N) \cdot C_i) = \underbrace{(P \cdot C_i)}_0 - (M_1 \cdot C_i) \leq 0$$

since M_1 is effective with no components of $\{C_i\}_{i=1,\dots,q}$. Then by effectiveness of B with components of $\{C_i\}_{i=1,\dots,q}$, we have:

$$((N_1 - N) \cdot B) \leq 0.$$

So all the inequalities above are equalities, and we have $(B^2) = 0$. By the negative definiteness of $\|(C_i \cdot C_j)\|_{i,j=1,\dots,q}$, we have $B = 0$, which proves the claim.

Finally,

$$D' - N = (N_1 - N) + M_1 \text{ is effective, hence } D' - N \in |P|.$$

Step 2: Now we prove the general case. It suffices to show the following map is surjective:

$$|mD - \lceil mN \rceil| \rightarrow |mD| : \quad E \mapsto E + \lceil mN \rceil.$$

For all $E \in |mD|$, choose $k \in \mathbb{N}^*$ such that mkP and mkN are integral divisors. $mkD = mkP + mkN$ is the Zariski decomposition of mkD . By Step 1, $kE - mkN$ is integral effective. Then:

$$E - \lceil mN \rceil = \lfloor \frac{kE - mkN}{k} \rfloor \text{ is still effective,}$$

hence $E - \lceil mN \rceil \in |mD - \lceil mN \rceil|$. □

Combining the above proposition with Fujita's vanishing theorem, we have the following

corollary.

Corollary 5.3. In the situation of Theorem 5.1,

$$\mathrm{vol}(D) = \mathrm{vol}(P) = (P^2).$$

In particular, the volume of an integral divisor on a surface is always a rational number.

Remark. Recall that we constructed a 3-dimensional smooth projective variety X with a big line bundle $\mathcal{L}$ such that $\mathrm{vol}(\mathcal{L})$ is irrational in Theorem 2.3. It is natural to find that we cannot always have a Zariski decomposition as above on higher dimensional varieties.

To introduce another corollary of Zariski decomposition, we need the following generalized Wilson's Theorem.

Lemma 5.4 (Generalized Wilson's Theorem). Let X be irreducible projective variety with $\dim X = n$. Let D be a nef and big $\mathbb{Q}$ -divisor on X . Then there exists $m_0 \in \mathbb{Z}_{>0}$ and an effective divisor $\tilde{N}$ such that: for all $m \geq m_0$, we have

$$|[mD] - \tilde{N}| \text{ base point free.}$$

Proof. Choose $k \in \mathbb{N}^*$ such that kD is an integral divisor and:

$$kD \equiv_{\mathrm{lin}} A + E \quad \text{with } A \text{ ample and } E \text{ effective.}$$

By Wilson's Theorem, there exists $m_0^* \in \mathbb{Z}_{>0}$ and an effective divisor N such that: for all $m \geq m_0^*$, we have $|mkD - N|$ base point free. Choose $j_0 \in \mathbb{N}^*$ such that:

$$j_0A + [rD] \text{ is very ample (hence free) for all } r \in [0, k).$$

Choose $m_0 = (j_0 + m_0^*) \cdot k$, $\tilde{N} = N + j_0E$. Then for all $m \geq m_0$, we can write $m = j \cdot k + r$ with $j \geq j_0 + m_0^*$ and $r \in [0, k)$. Then:

$$\begin{aligned} [mD] - \underbrace{(N + j_0E)}_{\text{effective}} &= jkD + [rD] - (N + j_0E) \\ &\equiv_{\mathrm{lin}} \underbrace{((j - j_0)kD - N)}_{\text{free}} + \underbrace{(j_0A + [rD])}_{\text{free}} \end{aligned}$$

hence free. □

Corollary 5.5. In the situation of Theorem 5.1, further assume that D is big. Given $m \gg 1$ denote by $\text{ord}_E(|mD|)$ the order along E of a general divisor in $|mD|$. Then the following limit exists and is a rational number:

$$\lim_{m \rightarrow \infty} \frac{\text{ord}_E(|mD|)}{m} = \text{ord}_E(N).$$

Proof. By the last corollary, $(P^2) = \text{vol}(D) > 0$, hence is a nef and big $\mathbb{Q}$ -divisor on X . By the previous proposition:

$$\text{ord}_E(|mD|) = \text{ord}_E(|\lfloor mP \rfloor|) + \text{ord}_E(\lceil mN \rceil) \quad \forall m \in \mathbb{N}.$$

Apply the generalized Wilson's Theorem to P , we have:

$$\text{ord}_E(|\lfloor mP \rfloor|) \leq \text{ord}_E(\tilde{N}) + \underbrace{\text{ord}_E(\lfloor mD \rfloor - \tilde{N})}_0 \quad \forall m \gg 1,$$

which is bounded by a constant independent of m . So we have:

$$\lim_{m \rightarrow \infty} \frac{\text{ord}_E(|mD|)}{m} = \text{ord}_E(N).$$

□

6 Proof of Zariski Decomposition

6.1 Sketch of proof of existence of Zariski Decomposition

First we give a sketch of the proof.

If D is nef, then we can take $P = D$ and $N = 0$. If D is not nef, we have the following lemma.

Lemma 6.1. D is a pseudo-effective divisor on a smooth projective surface X . Then there are finitely many irreducible curves $C_1, \dots, C_q$ on X such that $(D \cdot C_i) < 0$ for all $i = 1, \dots, q$. Especially, if $C_1, \dots, C_q$ are the only curves with negative intersection with D , then $q = 0$ if and only if D is nef.

Denote $C_1, \dots, C_{q_1}$ as the curves with negative intersection with D . We will show that:

$$\|(C_i \cdot C_j)\|_{i,j=1,\dots,q_1} \text{ is negative definite.} \quad (1)$$

The key lemma will be mentioned soon. Then we can find a unique effective divisor $N_1 = \sum_{i=1}^{q_1} a_i C_i$ such that:

$$(D - N_1) \cdot C_i = 0 \text{ for all } i = 1, \dots, q_1.$$

If $D_1 := D - N_1$ is nef, then we can take $P = D_1$ and $N = N_1$. The only thing to show is that:

$$N_1 \text{ is effective.} \quad (2)$$

If D_1 is not nef, we can repeat the same process to D_1 and find curves $C_{q_1+1}, \dots, C_{q_2}$ with negative intersection with D_1 . It should be noted that we need to show:

$$D_1 \text{ is pseudo-effective} \quad (3)$$

to apply Lemma 6.1 to D_1 .

$\|(C_i \cdot C_j)\|_{i,j=q_1+1,\dots,q_2}$ is negative definite by the same reason as (1), but we will not use it. Instead we will show that:

$$\|(C_i \cdot C_j)\|_{i,j=1,\dots,q_2} \text{ is negative definite.} \quad (4)$$

(1) and (4) come immediately from the following key lemma:

Lemma 6.2. Let $C_1, \dots, C_q$ be irreducible curves on a smooth projective surface X and D a pseudo-effective divisor on X . Assume that there exists $0 \leq r < q$ such that $(D \cdot C_i) < 0$ for all $i = r+1, \dots, q$ and $(D \cdot C_i) = 0$ for all $i = 1, \dots, r$. If $\|(C_i \cdot C_j)\|_{i,j=1,\dots,r}$ is negative definite, then $\|(C_i \cdot C_j)\|_{i,j=1,\dots,q}$ is negative definite.

Then we can find a unique effective divisor $N_2 = \sum_{i=1}^{q_2} a_i C_i$ such that:

$$(D_1 - N_2) \cdot C_i = 0 \text{ for all } i = 1, \dots, q_2.$$

N_2 is effective by the same reason as (2).

If $D_2 := D_1 - N_2$ is nef, then we can take $P = D_2$ and $N = N_1 + N_2$. If it is not nef, by the same reason as (3), D_2 is pseudo-effective. We can repeat the same process to D_2 and find curves $C_{q_2+1}, \dots, C_{q_3}$ with negative intersection with D_2 . By Lemma 6.2, we can show that $\|(C_i \cdot C_j)\|_{i,j=1,\dots,q_3}$ is negative definite. Then we can find a unique effective divisor $N_3 = \sum_{i=1}^{q_3} a_i C_i$ such that $(D_2 - N_3) \cdot C_i = 0$ for all $i = 1, \dots, q_3$. Discuss whether $D_3 := D_2 - N_3$ is nef or not, and repeat the same process.

The process will stop after finitely many steps because the negative definite property implies linear independence of the curves $C_1, \dots, C_{q_l}$ in $\text{NS}(X)_{\mathbb{R}}$ and $\dim \text{NS}(X)_{\mathbb{R}} < \infty$. So if the process stops at the l -th step, we can take $P = D_l$ and $N = N_1 + \dots + N_l$.

6.2 Proof of Lemma 6.2

Step 1: We first prove $\|(C_i \cdot C_j)\|_{i,j=1,\dots,q}$ is negative semi-definite by contradiction. Assume that there exists $Z \in \mathbb{Q}C_1 \oplus \dots \oplus \mathbb{Q}C_q$ such that $(Z \cdot Z) > 0$. By multiplying a positive integer, we can assume that Z is an integral divisor. We claim that we can also assume that Z is effective: This is because we can write $Z = Z_1 - Z_2$ where Z_1 and Z_2 are effective divisors with no common components. Then we have $0 < (Z \cdot Z) = (Z_1 \cdot Z_1) - 2(Z_1 \cdot Z_2) + (Z_2 \cdot Z_2)$. So at least one of $(Z_1 \cdot Z_1)$ and $(Z_2 \cdot Z_2)$ is positive. Then consider one of Z_1 and Z_2 .

Step 1.1: By Riemann-Roch Theorem on the surface X , we have:

$$h^0(nZ) + h^0(K_X - nZ) \geq \chi(nZ) = \frac{1}{2}(nZ \cdot nZ - nZ \cdot K_X) + \chi(\mathcal{O}_X).$$

Furthermore $h^0(K_X - nZ) = 0$ for $n \gg 0$:⁹ For any ample divisor H , we have $(Z \cdot H) > 0$ by effectiveness of Z . So for $n \gg 0$, $(K_X - nZ \cdot H) < 0$. But on the other hand, if $h^0(K_X - nZ) \neq 0$, $K_X - nZ \equiv_{\text{lin}} E$ for some effective divisor E . Then $(K_X - nZ \cdot H) = (E \cdot H) \geq 0$, which is a

⁹By the same reason we have a general consequence: If X is a projective surface and D is not pseudo-effective (which means there must be an ample divisor H such that $(D \cdot H) < 0$), then for any divisor F , $h^0(F + nD) = 0$ for $n \gg 0$.

contradiction. So we have:

$$h^0(nZ) \geq \frac{(Z \cdot Z)}{2}n^2 + O(n).$$

Step 1.2: On the other hand, we can use $D \in \overline{\text{Eff}}(X)$ to show:

$$h^0(nZ) = 1 \quad \text{for all } n \in \mathbb{N}.$$

Consider the fixed component of the linear system $|nZ|$, denoted by F_n . We can show that $Z_n := nZ - F_n$ is nef: This is because for any irreducible curve C on X , we can always find $E \in |Z_n|$ ¹⁰ such that C is not a component of E , otherwise contradicting the definition of F_n . Then we have $(Z_n \cdot C) = (E \cdot C) \geq 0$.

By the nefness of Z_n and the pseudo-effectiveness of D , we have $(D \cdot Z_n) \geq 0$. Notice that by definition $Z_n \in \mathbb{Z}C_1 \oplus \cdots \oplus \mathbb{Z}C_q$ and is effective, recall the condition that $(D \cdot C_i) = 0$ for $i = 1, \dots, r$ and $(D \cdot C_i) < 0$ for $i = r + 1, \dots, q$, so it can only be:

$$Z_n = \sum_{i=1}^r b_i C_i. \quad \text{for some } b_i \in \mathbb{N}.$$

Again by nefness of Z_n :

$$(Z_n \cdot Z_n) = \sum_{i=1}^r b_i (Z_n \cdot C_i) \geq 0.$$

By $\|(C_i \cdot C_j)\|_{i,j=1,\dots,r}$, it can only be $Z_n = 0$. So the fixed component $F_n = nZ$ and $|nZ|$ has only one element, which is nZ itself, which implies $h^0(nZ) = 1$ for all $n \in \mathbb{N}$.

Hence we have a contradiction to Step 1.1 and $(Z \cdot Z) > 0$. The assumption of Step 1 is false.

Step 2: We prove by induction on k that $\|(C_i \cdot C_j)\|_{i,j=1,\dots,r+k}$ is negative definite. $k = 0$ is the assumption of Lemma 6.2. If correct for k , we prove it for $k+1$. By Step 1, $\|(C_i \cdot C_j)\|_{i,j=1,\dots,r+k+1}$ is negative semi-definite. Prove by contradiction, suppose it is not negative definite, we can denote:

$$\begin{aligned} V &:= \mathbb{Q}C_1 \oplus \cdots \oplus \mathbb{Q}C_{r+k+1}, \\ V_1 &:= \mathbb{Q}C_1 \oplus \cdots \oplus \mathbb{Q}C_{r+k}, \\ V_2 &:= \{F \in V \mid (F \cdot F') = 0, \forall F' \in V\} = V_1^\perp. \end{aligned}$$

¹⁰A small discussion: $Z_n = 0$ then nothing to say; $Z_n \neq 0$ then $|Z_n|$ is non-empty because nZ is effective and by the definition of F_n , Z_n is effective.

Then:

$$V = V_1 \oplus V_2.$$

So we can write

$$C_{r+k+1} = -D_{r+k+1} + E_{r+k+1} \quad \text{for some } D_{r+k+1} \in V_1 \text{ and } E_{r+k+1} \in V_2.$$

Step 2.1: Now we claim that D_{r+k+1} is effective. Write:

$$D_{r+k+1} = A_{r+k+1} - B_{r+k+1} \quad \text{for some effective divisors } A_{r+k+1}, B_{r+k+1} \in V_1 \text{ with no common components.}$$

On one hand:

$$(D_{r+k+1} \cdot B_{r+k+1}) = (A_{r+k+1} \cdot B_{r+k+1}) - (B_{r+k+1} \cdot B_{r+k+1}) \geq 0$$

where $(B_{r+k+1} \cdot B_{r+k+1}) \leq 0$ by the negative definiteness of $\|(C_i \cdot C_j)\|_{i,j=1,\dots,r+k}$. On the other hand, note that:

$$(D_{r+k+1} \cdot C_i) = (E_{r+k+1} \cdot C_i) - (C_{r+k+1} \cdot C_i) = -(C_{r+k+1} \cdot C_i) \leq 0 \quad \text{for all } i = 1, \dots, r+k.$$

By the effectiveness of B_{r+k+1} with components of $\{C_i\}_{i=1,\dots,r+k}$, we have:

$$(D_{r+k+1} \cdot B_{r+k+1}) \leq 0.$$

So all the inequalities above are equalities, which implies $(B_{r+k+1} \cdot B_{r+k+1}) = 0$. By the negative definiteness of $\|(C_i \cdot C_j)\|_{i,j=1,\dots,r+k}$, we have $B_{r+k+1} = 0$, which implies D_{r+k+1} is effective.

Step 2.2: So $E_{r+k+1} = D_{r+k+1} + C_{r+k+1}$ is effective with components of $\{C_i\}_{i=1,\dots,r+k+1}$ and:

$$(E_{r+k+1} \cdot C_i) = 0 \quad \text{for all } i = 1, \dots, r+k+1$$

by the definition of V_2 . So E_{r+k+1} is nef and by the pseudo-effectiveness of D , we have

$$(D \cdot E_{r+k+1}) \geq 0.$$

But on the other hand, by the condition of Lemma 6.2 and effectiveness of D_{r+k+1} , we have:

$$(D \cdot E_{r+k+1}) = (D \cdot D_{r+k+1}) + (D \cdot C_{r+k+1}) \leq (D \cdot C_{r+k+1}) < 0,$$

a contradiction to the assumption. So $\|(C_i \cdot C_j)\|_{i,j=1,\dots,r+k+1}$ is negative definite, induction is complete, and the proof of Lemma 6.2 is complete. $\square$

6.3 Completion of the proof of existence

Proof of Lemma 6.1: Suppose for any $q \in \mathbb{N}$, we can find q irreducible curves $C_1, \dots, C_q$ on X such that $(D \cdot C_i) < 0$ for all $i = 1, \dots, q$. Then by Lemma 6.2, we have $\|(C_i \cdot C_j)\|_{i,j=1,\dots,q}$ is negative definite. Especially, $[C_1], \dots, [C_q]$ are linearly independent in $\text{NS}(X)_{\mathbb{Q}}$, which contradicts the fact that $\dim \text{NS}(X)_{\mathbb{Q}} < \infty$.

Proof of (2): Write $N_1 := A_1 - B_1$ where A_1, B_1 are effective divisors with no common components. On one hand:

$$(N_1 \cdot B_1) = (A_1 \cdot B_1) - (B_1 \cdot B_1) \geq 0$$

where $(B_1 \cdot B_1) \leq 0$ by the negative definiteness of $\|(C_i \cdot C_j)\|_{i,j=1,\dots,q_1}$. On the other hand, note that $(N_1 \cdot C_i) = (D \cdot C_i) < 0$ for all $i = 1, \dots, q_1$. By the effectiveness of B_1 with components of $\{C_i\}_{i=1,\dots,q_1}$, we have:

$$(N_1 \cdot B_1) \leq 0.$$

So all the inequalities above are equalities, which implies $(B_1 \cdot B_1) = 0$. By the negative definiteness of $\|(C_i \cdot C_j)\|_{i,j=1,\dots,q_1}$, we have $B_1 = 0$, which implies N_1 is effective.

Proof of (3): For any nef divisor H on X , by the negative definiteness of $\|(C_i \cdot C_j)\|_{i,j=1,\dots,q_1}$, we can find unique $Y \in \mathbb{Q}C_1 \oplus \dots \oplus \mathbb{Q}C_{q_1}$ such that:

$$(Y \cdot C_i) = -(H \cdot C_i) \leq 0 \quad \text{for all } i = 1, \dots, q_1.$$

By the same reason as the proof of (2), we can show that Y is effective.

Notice that for any $i = 1, \dots, q_1$, we have $((H + Y) \cdot C_i) = 0$; for any irreducible curve C on X which is not one of $\{C_i\}_{i=1,\dots,q_1}$, by nefness of H and effectiveness of Y , we have

$$((H + Y) \cdot C) = (H \cdot C) + (Y \cdot C) \geq 0.$$

So $H + Y$ is nef.

By the pseudo-effectiveness of D , we have:

$$(D_1 \cdot H) = (D \cdot H) - (N_1 \cdot H) = (D \cdot H) + (N_1 \cdot Y) = (D \cdot H) + (D \cdot Y) = (D \cdot (H + Y)) \geq 0.$$

By the arbitrariness of H , we have D_1 is pseudo-effective.

6.4 Proof of uniqueness of Zariski Decomposition

Write $D = P + N = P' + N'$ as two Zariski decompositions of D . $\{C_i\}_{i=1,\dots,q}$ are the irreducible components of N . Write

$$N' = A + B \quad \text{where } A \text{ is effective with components of } \{C_i\}_{i=1,\dots,q}, \\ \text{and } B \text{ is effective with no components of } \{C_i\}_{i=1,\dots,q}.$$

We claim that $A - N$ is effective. Write $N - A = \widehat{A} - \widehat{B}$ where $\widehat{A}$ and $\widehat{B}$ are effective divisors with no common components. On one hand:

$$((N - A) \cdot \widehat{A}) = (\widehat{A} \cdot \widehat{A}) - (\widehat{B} \cdot \widehat{A}) \leq 0$$

where $(\widehat{A} \cdot \widehat{A}) \leq 0$ by the negative definiteness of $\|(C_i \cdot C_j)\|_{i,j=1,\dots,q}$. On the other hand, note that:

$$\begin{aligned} ((N - A) \cdot C_i) &= ((N - N') \cdot C_i) + (B \cdot C_i) \\ &= (P' \cdot C_i) - \underbrace{(P \cdot C_i)}_0 + (B \cdot C_i) \\ &\geq 0 \quad \text{for all } i = 1, \dots, q \end{aligned}$$

by the nefness of P' and the effectiveness of B with no components of $\{C_i\}_{i=1,\dots,q}$. So by the effectiveness of $\widehat{A}$ with components of $\{C_i\}_{i=1,\dots,q}$, we have:

$$((N - A) \cdot \widehat{A}) \geq 0.$$

So all the inequalities above are equalities, which implies $(\widehat{A} \cdot \widehat{A}) = 0$. By the negative definiteness of $\|(C_i \cdot C_j)\|_{i,j=1,\dots,q}$, we have $\widehat{A} = 0$, which implies $A - N$ is effective.

So $N' - N = (A - N) + B$ is effective. By the same reason, we can show that $N - N'$ is effective. So $N = N'$ and $P = P'$.